

SATYAM PATIL

Maharashtra, India

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EDUCATION

Indian Institute of Technology, Mandi

B.Tech, Computer Science and Engineering (CGPA of 8.25)

Nov 2021 – Jun 2025

Himachal Pradesh, India

TU Dresden

Semester Exchange Student in Computer Science and Engineering (SGPA of 9)

Oct 2023 – Feb 2024

Dresden, Germany

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, JavaScript, SQL, Kotlin, Dart

AI Engineering: LLM agents (ReAct loops, tool calling, multi-agent orchestration), RAG pipelines, FAISS, FastEmbed, LangGraph, MCP, knowledge graphs (Neo4j, Cypher), OpenAI / Anthropic / Gemini APIs

Backend & Data: FastAPI, PostgreSQL, async Python, Flask, NodeJS, ReactJS, VueJS, React-Native, Pandas, NumPy, scikit-learn, Git, Docker

Systems: Android / AOSP (Soong, Android.bp), SEAndroid & SELinux policy, TEE-Rich OS interfaces, Linux

EXPERIENCE

Samsung Research

June 2025 – Present

Research Engineer | LLM Agents, RAG, Android Security, Automation

Noida

- Built **AX_Agent**, an in-house **ReAct-loop agentic system** running fully on internal infrastructure, exposing internal security and porting scripts as agent tools, letting engineers drive multi-step porting and policy tasks in natural language on a confidential codebase external LLMs cannot touch.
- Built **CoreBrain**, a **RAG system over internal JIRA and project-lifecycle issues plus their resolution commits**, summarizing each into a problem/solution pair so a newly filed issue retrieves its closest prior fixes and a resolution path.
- Engineered an **LLM-driven SEAndroid automation pipeline** that classifies incoming policy issues, generates and validates candidate SELinux rules, and prepares change lists, automating roughly **80%** of previously manual policy debugging.
- Resolved **100+ SEAndroid and security module issues** (policy misconfigurations, access violations, system-call denials); owned the **ICCC module** (TEE ↔ Rich OS communication), contributing to both the trusted app and its Android-side client service.

Brainwave Science

Feb 2024 – Dec 2024

Full Stack Developer Intern | VueJS, PostgreSQL, Fastify, PsychoPy, React-Native

Remote

- Built a unified **full-stack platform**: a VueJS frame for browser-based PsychoPy experiments, an RTSP IP-camera streaming system, and a cross-platform educational app (Android/iOS) for **500+ users**.

LASR Lab, TU Dresden

Oct 2023 – Jan 2024

Application Development Intern | Kotlin, Gradle, Machine Learning, Computer Vision

Dresden, Germany

- Built an **Android application** for the Digit tactile sensor with **real-time on-device object detection**, cutting false detection rates by **30%**.

PROJECTS

Rankit – AI Mock-Test Platform (Co-founder) | FastAPI, PostgreSQL, LangGraph, FAISS, OpenAI API | rankit.in

- Own the full stack of a **live JEE mock-test platform** used by real students (FastAPI, PostgreSQL, Vite frontend, and the AI analysis layer), deployed across Vercel, Fly.io, and Neon.
- Built a **multi-agent analysis backend** with query routing, task decomposition, and tool selection: post-test SQL aggregates feed an LLM that pinpoints weak topics and generates personalized practice sets, backed by a **knowledge-graph layer** (topics as nodes, prerequisites as edges) that traces root-cause gaps instead of flat topic similarity.
- Built a **vision-LLM question-bank pipeline** classifying scanned questions into a deduplicated chapter/topic taxonomy and parsing PYQ documents, including Word equations.

CLAI – Context-Aware CLI Agent | Python, FAISS, FastEmbed, SQLite, Gemini API | [Github](https://github.com)

- Built a terminal agent turning natural language into shell commands, backed by **persistent long-term memory** (SQLite + FAISS/FastEmbed retrieval over past commands, folders, and workflows) and a confirmation gate on destructive operations.

ACHIEVEMENTS & POSITIONS OF RESPONSIBILITY

- World's top 21 teams**, CANSAT'23 (NASA / AAS) | **India's top 12**, SIF Space Hackathon finalist (ISRO / IISF).
- 1st Position**, Blockchain Hackathon, Cognizance'24 IIT Roorkee | **Best Blockchain Project**, Frosthack'23 IIT Mandi | **1st Position**, Astrophysics Hackathon (Merope).
- Co-coordinator**, STAC IIT Mandi | **Media & Outreach Head**, Xpecto'23 | **Core Team**, E-Cell IIT Mandi.